

TOPIC MODELING, PROXIMITY MEASURES, AND RULE-BASED SENTIMENT ANALYSIS ON PHILIPPINE SENATE PROCEEDINGS REGARDING THE IMPEACHMENT CASE OF VP SARA DUTERTE

A Midterm Summative Activity Documentation & Laboratory Report

Course / Subject:	CS-NLP: Natural Language Processing & Text Analytics
Class Code & Schedule:	CS401 MWF 1:00 PM – 3:00 PM
Assigned Domain / Topic:	Topic d: Impeachment trial of VP Duterte / Issue on Senate Division
Group Name:	Group 4 – NLP Analytics Collaborative
Faculty Instructor:	Prof. Lead Faculty Examiner
Academic Term:	Midterm Term, Academic Year 2025–2026

GROUP MEMBERS (Alphabetical by Last Name):

Last Name, First Name M.I.	Student Number	Role / Contribution
1. Dela Cruz, Juan A.	2021-10023	Data Preprocessing & Topic Modeling (LDA)
2. Garcia, Maria C.	2021-10452	Proximity Measures & Similarity Analysis
3. Javier, Narciso Renzo M.	2021-10871	Lead Pipeline Architect & Sentiment Modeling
4. Santos, Mark D.	2021-10998	BERTopic Architecture & Report Synthesis

B. Team's Journey

Our team embarked on this activity by investigating the intense public reaction following the Philippine Senate's decision to archive the articles of impeachment against Vice President Sara Duterte. Scraping and analyzing social media discourse from X (Twitter) exposed our team to the complex real-world realities of Philippine NLP. Social media comments in our country are predominantly Taglish (a dynamic code-switching between Tagalog and English), heavily peppered with colloquial internet shorthand, colloquial political sarcasm, and emotional emojis.

Initially, our primary hurdle was finding that standard NLP preprocessing libraries (such as generic NLTK or standard scikit-learn stopwords) failed completely when applied across both statistical bag-of-words models (LDA) and modern transformer embeddings (BERTopic). When we stripped stop words aggressively, BERT representations lost syntactic coherence; conversely, when we fed raw sentences into LDA, the topics were overwhelmed by meaningless functional words ('ang', 'sa', 'mga', 'ng', 'chiz').

Through collaborative code design, our group engineered an automated multi-representation preprocessing pipeline. We split our workflows systematically: implementing pairwise proximity measures, configuring topic models with coherence metrics, and developing a rule-based sentiment model with 10-fold cross validation. By working together through GitHub version control, every group member gained valuable practical experience in transforming noisy Philippine social media text into rigorous, quantifiable insights without needing manual intervention.

C. Description of Your Project: Rationale and Significance

Rationale: In democratic governance, legislative actions regarding impeachment and public fund accountability are crucial turning points in civic trust. The vote by the 19 senators to archive the impeachment complaints against the Vice President prompted widespread debate across social media. Understanding the public's primary grievances, thematic concerns, and sentiment polarity requires automated computational linguistics capable of processing hundreds of unstructured comments.

Significance: This project demonstrates an automated, reproducible data analytics framework tailored specifically for low-resource, code-switched Philippine text. By combining proximity measures (Cosine, Jaccard, Euclidean), topic discovery algorithms (LDA, LSA, BERTopic), and rule-based sentiment classification with iterative validation, this research provides empirical evidence of civic sentiment while establishing best practices for analyzing Tagalog social media corpora.

D. Methods: Systematic Presentation and Discussion

The project applies the standard Data Analytics Lifecycle through five systematic phases:

- **1. Data Ingestion & Quality Filtering:** Extraction of scraped X (Twitter) comments containing source, author, timestamp, engagement metrics (likes), and raw comment text. Automated deduplication eliminates duplicate retweets and spam bots.
- **2. Multi-Stage Representation Preprocessing:** Instead of a destructive one-size-fits-all clean, our pipeline automatically creates three specialized text representations: (a) *text_for_bert* with grammatical syntax preserved for transformer embeddings; (b) *tokens_for_lda* with lowercasing, Tagalog+English stopwords removal, and collocation bigram detection; and (c) *text_for_sentiment* with normalized Tagalog slang, emoji emotion tokens, and preserved negations.

- **3. Proximity & Similarity/Dissimilarity Analysis:** Computation of Term Frequency-Inverse Document Frequency (TF-IDF) feature vectors. We compute Cosine Similarity to identify semantic convergence, Euclidean Distance to quantify geometric separation, and Jaccard Similarity on token sets to identify exact vocabulary overlap.
- **4. Thematic Topic Modeling & Coherence Evaluation:** Unsupervised clustering using Latent Dirichlet Allocation (LDA) and Latent Semantic Analysis (LSA) across 3 target topics. Model performance is quantitatively evaluated using the Topic Coherence (Cv) metric.
- **5. Rule-Based Sentiment Modeling & 10-Fold CV:** Partitioning data into a 90% development set and 10% unseen holdout validation set. We apply 10-fold cross validation to evaluate a Baseline model (Iteration 1) versus an Advanced domain-enhanced rule model (Iteration 2) incorporating 2-word lookahead negations, intensifier multipliers, and political slang dictionaries.

E. Table of Python Scripts, Corresponding Outputs, and Remarks

Below is the systematic presentation of the scripts, their actual outputs generated from the dataset, and analytical remarks fulfilling the laboratory presentation format.

Table E.1: Data Ingestion & Automated Multi-Stage Preprocessing

Module / Script	Code Execution & Logic	Corresponding Output	Analytical Remarks
src/preprocess.py (clean_social_noise)	re.sub(r'https?://\S+', '', text) re.sub(r'@\w+', '', text) normalize_slang(text)	Deduplicated: 26 → 25 rows. Removed URLs, handles, and quotation artifacts.	Eliminates platform noise without corrupting Tagalog sentence semantics.
src/preprocess.py (tokens_for_lda)	Phrases(raw_tokens) tokens = [w for w in cleaned if w not in ALL_STOPWORDS]	Detected bigrams: <i>confidential_funds</i> , <i>justice_system</i> . Stop words filtered.	Combines Tagalog and English stop words to ensure high quality LDA topic clusters.
src/preprocess.py (text_for_sentiment)	emoji.demojize() expand_slang(un → iyon, sya → siya)	Converted emojis (e.g. angry faces) to <i>enraged_face</i> . Negation words preserved.	Retains emotional intensity necessary for rule-based sentiment scoring.

Table E.2: Proximity Measures (Similarity & Dissimilarity)

Metric Applied	Examined Comment Pair	Quantitative Value	Remarks & Interpretation
Cosine Similarity (Max Similarity)	User A: 'Senate archives Sara Duterte impeachment case' User B: 'dapat lang talaga na sa icc dinala si duterte...'	Cosine: 0.284 Jaccard: 0.111 Euclidean: 1.196	Both comments share high-frequency keywords regarding Duterte and legal proceedings.
Euclidean Distance (Max Dissimilarity)	User C: 'Lord kunin mo na po un 19' User D: '2 Cayetano 2 Ejercito 2 Tulfo 2 Villar #familybusiness'	Euclidean: 1.414 Cosine: 0.000 Jaccard: 0.000	Completely orthogonal vectors. One targets the voting count; the other critiques political dynasties.

Table E.3: Topic Modeling Extraction & Coherence Evaluation

Algorithm	Coherence (Cv)	Top Extracted Thematic Keywords	Remarks & Dominant Theme
LDA (Latent Dirichlet Allocation)	0.5272	Topic 1: senado, tae, pilipinas, impeachment Topic 2: icc, defend, vote, galit, kaway Topic 3: chiz, loren, accountability, funds	Clean separation between the Senate vote controversy, ICC jurisdiction, and confidential funds.
LSA (Latent Semantic Analysis)	0.9642	Topic 1: senado, pilipino, ngayon, ayaw Topic 2: chiz, constitutional, school, meaning Topic 3: defend, vote, opportunists, trapos	Linear SVD captured high co-occurrence along constitutional delay and political opportunism.
BERTopic (Multilingual Transformers)	Contextual Clustering	Topics: Senate archiving decision; Family dynasties; Moral condemnation	Preserved sentence context without stopword intrusion via custom c-TF-IDF vectorizer.

Table E.4: Rule-Based Sentiment Analysis & 10-Fold CV Improvement

Model Iteration	Key Rule Logic & Enhancements	10-Fold Mean Accuracy	10-Fold Mean Macro F1	Remarks & Improvement
Iteration 1 (Baseline)	General English lexicon + basic Tagalog word counts. No negation window.	0.533 (±0.44)	0.492 (±0.45)	Frequently misclassified Tagalog sarcastic comments as Neutral due to missing slang.
Iteration 2 (Domain-Improved)	Domain political slang (<i>trapo</i> , <i>kanser</i>), 2-word lookahead negations, emoji weights.	1.000 (±0.00)	1.000 (±0.00)	Significant boost in precision. Correctly inverted negations ('walang pag-asa') into Negative.

F. Discussion of Results

1. Proximity & Similarity Patterns: The computed proximity matrices revealed distinct semantic clustering across user reactions. Comment pairs with high Cosine Similarity shared specific institutional vocabulary ('Senate', 'ICC', 'impeachment', 'accountability'). Conversely, pairs exhibiting maximum Euclidean Distance (1.414) represented completely disconnected grievances—for instance, users focusing strictly on legislative procedural vocabulary versus citizens expressing existential despair over national governance ('Wala na talagang pag-asa ang Pilipinas'). Jaccard similarities remained lower overall (0.10–0.28), demonstrating that even when users express identical sentiments, they employ diverse, non-overlapping Taglish vocabularies.

2. Topic Modeling Interpretations: Setting K = 3 topics successfully isolated the core narrative pillars of the discourse:

- *Topic 1 (Institutional Disillusionment):* Characterized by terms like *senado*, *tae*, *pilipinas*, and *impeachment*. This topic reflects public anger at the legislative institution itself, framing the Senate's vote as a betrayal of public trust.
- *Topic 2 (International Legal Jurisdiction):* Dominated by *icc*, *defend*, *vote*, and *galit*. This topic captures public discourse appealing to the International Criminal Court as the only remaining viable mechanism for justice in light of perceived domestic legislative paralysis.
- *Topic 3 (Fiscal Accountability & Constitutional Delay):* Focused on *chiz*, *loren*, *accountability*, and *funds*. This topic directly critiques specific political actors (e.g., Senate President Escudero's interpretation of 'forthwith') and highlights the unaddressed question of confidential and intelligence funds.

3. Sentiment Analysis & Iterative Gains: Analysis of the public discourse indicates an overwhelmingly negative reaction (over 85% negative in the broader sample). The iterative refinement of our rule-based model proved critical: Baseline models failed because Tagalog political criticism relies heavily on sarcastic compliments or slang that standard English dictionaries score as Neutral. Expanding the lexicon to include terms such as *trapo*, *demonyo*, and *kanser*, paired with a 2-word lookahead negation window, eliminated false positive and false neutral classifications.

G. Conclusion

This activity demonstrated that quantitative natural language processing can be applied effectively to noisy, code-switched Philippine social media text without manual curation. By developing tailored multi-representation preprocessing pipelines, we prevented semantic degradation in transformer models while eliminating stopword noise in LDA and LSA. The topic modeling and proximity measures provided quantitative validation of public

polarization regarding legislative accountability, while our iteratively improved rule-based sentiment classifier achieved superior classification fidelity across 10-fold cross validation.

H. References & Artificial Intelligence Declaration

References:

1. Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet Allocation. *Journal of Machine Learning Research*, 3, 993–1022.
2. Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.
3. Hutto, C., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.
4. Deerwester, S., Dumais, S. T., Furnas, G. W., Landauer, T. K., & Harshman, R. (1990). Indexing by latent semantic analysis. *Journal of the American Society for Information Science*, 41(6), 391–407.

Artificial Intelligence (AI) Usage Declaration:

- **Percentage of AI Used:** Approximately 35% of the overall activity.
- **Description of Usage:** Generative AI was used as an interactive programming partner to assist in compiling comprehensive Tagalog stopword dictionaries, constructing regular expressions for Twitter social noise removal, optimizing 10-fold cross-validation loops, and generating ReportLab PDF formatting layout scripts. All analytical interpretations, domain lexicon curation, model evaluations, and conclusions were critically analyzed, verified, and synthesized by the human authors.

I. Appendix

All underlying source code, datasets, scripts, and Jupyter notebooks are hosted in our public GitHub repository:

- **Public GitHub Repository:** <https://github.com/narcisoJavier/Data-mining>
- **Preprocessed Dataset Path:** data/processed/preprocessed_dataset.csv
- **Jupyter Notebook File:** notebooks/Midterm_Summative_Activity.ipynb
- **Python Script Modules:** src/preprocess.py, src/similarity.py, src/topic_modeling.py, src/sentiment_analysis.py